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```python
import heapq

def huffman_coding(characters, frequencies):

    heap = []

    for char, freq in zip(characters, frequencies):
        heapq.heappush(heap, (freq, char))

    while len(heap) > 1:

        freq1, left = heapq.heappop(heap)
        freq2, right = heapq.heappop(heap)

        new_node = (left, right)

        heapq.heappush(heap, (freq1 + freq2, new_node))

    root = heap[0][1]

    codes = {}

    def generate_codes(node, code=""):

        if isinstance(node, str):
            codes[node] = code
            return

        left, right = node

        generate_codes(left, code + "0")
        generate_codes(right, code + "1")

    generate_codes(root)

    return codes


characters = ['A', 'B', 'C', 'D']
frequencies = [5, 9, 12, 13]

codes = huffman_coding(characters, frequencies)

print("Huffman Codes:")

for char in characters:
    print(char, ":", codes[char])
```

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```
Python 3.14.3 (tags/v3.14.3:323c59a,
on win32
Enter "help" below or click "Help" a
>>>
==== RESTART: C:/Users/ROOPA SAI DIV
Huffman Codes:
A : 00
B : 01
C : 10
D : 11
>>>
```